Lab: Living Prediction Machines -- From Bacteria to Humans

Objective

Investigate how **biological agents** at different complexity levels (bacteria, plants, animals, humans) maintain preferred states and minimize surprise, connecting cellular mechanisms to the Active Inference framework.

Prerequisites

- Completed the Biology & Health Systems module
- Read the module on Living Things as Prediction Machines

Part 1: Agent Complexity Spectrum

Goal: Compare biological agents across a complexity gradient.

Research and fill in the table for each organism:

Feature	E. coli (bacterium)	Sunflower (plant)	Mouse (mammal)	Human
Preferred states				
Sensory mechanisms				
Actions available				
Model complexity (simple/ moderate/complex)				
Can it learn from experience?				
Does it have a nervous system?				

Write your response here

Part 2: Chemotaxis -- The Simplest Agent

Goal: Understand bacterial chemotaxis as the most basic form of Active Inference.

E. coli bacteria swim toward nutrients and away from toxins using a process called chemotaxis. Research this process and answer:

1. What is the bacterium's "preferred state"? (Hint: high nutrient concentration.)
2. How does it "sense" the environment? (Receptor proteins on its surface.)
3. What are its two possible actions? (Run straight or tumble randomly.)
4. How does it "decide" which action to take? (Compare current vs. recent chemical concentration.)

5. Is this a generative model? The bacterium compares "expected" concentration to "actual" concentration -- what is this comparison called in Active Inference?

Write your response here

Part 3: Your Body's Hidden Agents

Goal: Identify subsystems within your body that act as agents.

Your body contains many semi-autonomous agents. For each, describe its agency:

Subsystem	What It Maintains	How It Senses	How It Acts	Can It Learn?
Immune system				
Gut microbiome				
Autonomic nervous system				
Endocrine system				

Write your response here

Part 4: Disease as Agent Failure

Goal: Analyze a disease or disorder as a failure of biological agency.

Choose one condition (diabetes, autoimmune disease, anxiety disorder, or another you know about). Map it:

1. What preferred state is the body failing to maintain?
2. What is wrong with the sensory mechanism?
3. What is wrong with the action mechanism?
4. What is wrong with the internal model?
5. How do treatments attempt to restore proper agency?

Write your response here

Summary Table

Concept	Biological Example	Active Inference Connection
Biological Agent	Any organism that maintains preferred states	Free energy minimization through action
Chemotaxis	Bacterial movement toward nutrients	Simplest form of Active Inference
Nested Agency	Immune cells within a body	Agents within agents
Agent Failure	Disease states	Breakdown of prediction-action loops